Frequentist, Bayesian and machine-learning analysis of the stiffness of resistant breast-cancer cells from AFM force curves

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We analyse 257 AFM force–distance curves measured on the cytoplasm and on the nucleus of two resistant breast-carcinoma lines (MCF7R and T47DR). Each curve is reduced to the effective elastic parameter A of a Hertz-like model, $F = A\,\delta^{3/2}$, and the distributions of $\log A$ are compared between regions with the same statistical toolbox in a frequentist (profile likelihood, toy Monte Carlo, Holm correction, systematics), a Bayesian (MCMC, Bayes factors, hierarchical model) and a machine-learning (anomaly detection with injected defects, classification, mixtures) setting. The nucleus is softer than the cytoplasm in both lines: $\Delta = \mu_{nucleus} - \mu_{cytoplasm} = -0.81 \pm 0.08$ (MCF7R) and -0.25 ± 0.10 (T47DR) in $\log A$, i.e. a ratio of typical A of 0.44 and 0.78. The evidence is overwhelming for MCF7R and only weak for T47DR with the fit parameter (Bayes factor $\sim$ 4:1 despite $p \simeq 0.01$), and model-independent checks (force at fixed indentation, indentation at fixed force) give the same sign in both lines at every depth. The acquisition timestamps and calibration metadata show, however, that cytoplasm and nucleus were measured in separate, consecutive time blocks, that the first curve at a spot is stiffer than the following ones, and that the stored spring constant of the same cantilever differs by 40% between the two sessions. A single hierarchical model including a temporal drift, the position in the burst of consecutive curves and a burst effect keeps the MCF7R result ($\beta = -0.88$, 95% interval $[-1.22, -0.55]$) but leaves the T47DR stiffness difference undetermined ($+0.03$, $[-0.35, +0.41]$), while the higher contact-law exponent of the T47DR nucleus is robust. Differences between lines cannot be interpreted, and the analysis is also limited by pseudo-replication.

I. INTRODUCTION

Cell stiffness measured by atomic force microscopy (AFM) is a label-free marker of cell state, and of the response of tumour cells to treatment. The aim of this work is to compare the mechanical properties of two resistant breast-carcinoma lines, MCF7R and T47DR, in two measurement regions (cytoplasm and nucleus) at 48 h, starting from raw force–distance curves. The emphasis is on the *statistical* treatment: every comparison is made with explicit models, its behaviour is validated with toy Monte Carlo experiments, and the dependence on analysis choices is quantified. The pipeline is implemented in ten notebooks (reading and QC, contact point and fit, frequentist, Bayesian and machine-learning analysis, depth-resolved stiffness, acquisition time, calibration and a unified model; see Sec. X).

II. DATA AND CURVE REDUCTION

The data are 257 JPK force files in four groups: MCF7R cytoplasm (55 curves), MCF7R nucleus (60), T47DR cytoplasm (59) and T47DR nucleus (83), measured with an MLCT-BIO triangular cantilever (spring constant $\simeq 0.12$ N/m). Each file holds an extend (approach) and a retract segment; the extend segment is used for the mechanical fit, the retract one for the adhesion. The curves are the «good curves» already selected by the experimenters; a preliminary instrumental QC (robust z-scores of noise, force excursion and jumps) is computed but *no curve is discarded*.

For each extend curve the baseline is a robust line fitted on the first 30% of the points (the tip is far from the surface), the noise σ_b is the MAD of its residuals, and the contact point is initialised where the smoothed force exceeds $\max(8\sigma_b,\, 5\%F_{max})$ for 20 consecutive points. In the window $0.08F_{max} < F < 0.8F_{max}$ the data are fitted with

$$F(\delta) = A\,\delta^{3/2}, \qquad \delta = z_c - z, \tag{1}$$

with A and the contact point z_c free [1]. The tip radius and the Poisson ratio are not available, so A is an *effective* parameter in $\mathrm{nN}\,\mu\mathrm{m}^{-3/2}$ and not the Young modulus; the nominal tip geometry is pyramidal, for which a Sneddon-type law [2] would have exponent 2, but the free exponent measured on the curves has median 1.47, consistent with the 3/2 used here; in any case only *relative* comparisons are meaningful. The fits have median $R^2 = 0.986$ and a relative statistical error on A of 0.8%, far smaller than the spread of A between curves (Sec. IV), which is therefore intrinsic. Figure 1 shows one fit. A is positive and right-skewed, so the analysis uses $\log A$ (with A in $\mathrm{nN}\,\mu\mathrm{m}^{-3/2}$).

III. STATISTICAL MODEL

In each group g we model $x_i = \log A_i \sim \mathcal{N}(\mu_g, \sigma_g^2)$ (a log-normal A). Figure 2 shows the distributions: the Gaussian is adequate for the bulk, and $\log A$ is much more symmetric than A (skewness of A from 0.8 to 3.1, of $\log A$ from -0.2 to 0.9; the largest values are in the nucleus groups, see Sec. V). The parameter of interest is the nucleus–cytoplasm shift $\Delta_l = \mu_{l,nucleus} - \mu_{l,cytoplasm}$ for each line l; e^{Δ} is the ratio of the typical A, and $\Delta < 0$ means a softer nucleus.

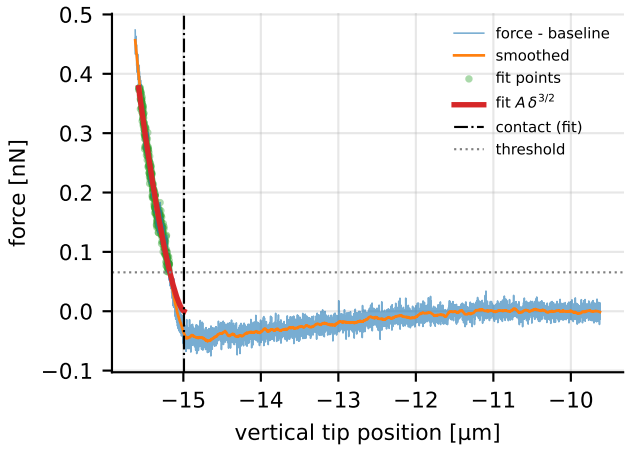


FIG. 1. Example of baseline-corrected curve, contact point and Hertz-like fit (MCF7R cytoplasm).

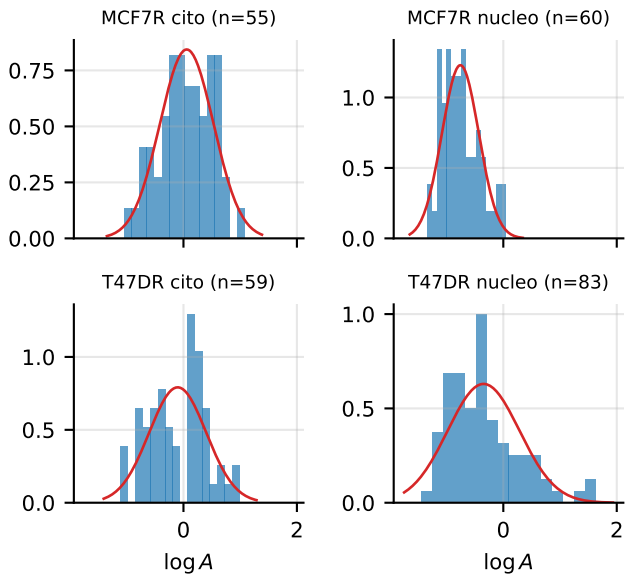


FIG. 2. Distribution of $\log A$ in the four groups with the fitted Gaussian.

IV. FREQUENTIST ANALYSIS

Estimation. The parameters are obtained from the unbinned likelihood with `iminuit` [3]; Hesse and Minos errors coincide with the analytic $\sigma/\sqrt{n}$ (e.g. MCF7R nucleus: $\hat{\mu} = -0.757 \pm 0.042$, $\hat{\sigma} = 0.325 \pm 0.030$).

Profile likelihood. For fixed Δ the maximisation over the other parameters is analytic up to $\mu_{cytoplasm}$, so the profile is computed exactly. The 68% intervals and the Wilks test [4] give $\Delta_{\mathrm{MCF7R}} = -0.814\,^{+0.077}_{-0.076}$ ($q = 70.7$, $p < 10^{-15}$) and $\Delta_{\mathrm{T47DR}} = -0.247\,^{+0.096}_{-0.095}$ ($q = 6.5$, $p = 0.011$). Welch, Mann–Whitney, permutation and bootstrap-of-medians tests agree (T47DR: $p = 0.011$, 0.003, 0.015; median difference -0.27, 95% interval $[-0.67, -0.07]$), so the conclusion does not depend on the Gaussian assumption.

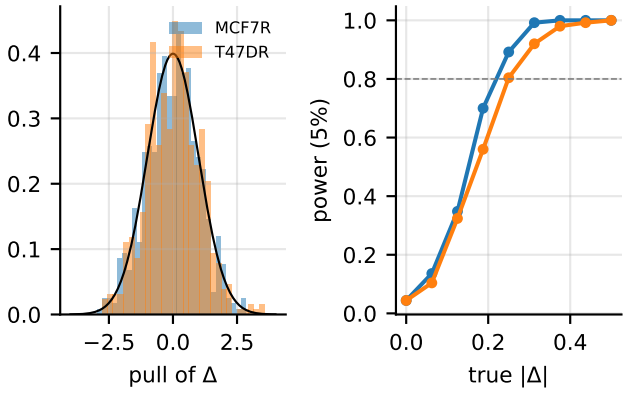


FIG. 3. Toy Monte Carlo of the profile-likelihood estimator. Left: pull of Δ (black: standard normal). Right: power of the Wilks test versus the true $|\Delta|$.

Toy Monte Carlo validation. Generating 1500 experiments per line from the fitted model, the estimator is unbiased (bias $< 10^{-3}$), the pulls have mean 0.01 and width 0.98–1.00, the 68% intervals cover 0.680 and 0.679, and the size at the nominal 5% is 0.059 (MCF7R) and 0.052 (T47DR), equal to Welch and Mann–Whitney. The power at the real sample size is 1.00 for MCF7R and 0.73 for T47DR; 80% power requires $|\Delta| \gtrsim 0.22$–0.25, so T47DR sits just below the sensitivity threshold (Fig. 3).

Interaction and multiple tests. A nested test of the additive model $\mu_{l,r} = a_l + b_r$ against the full model rejects it ($q = 20.6$, $p = 6 \cdot 10^{-6}$): the nucleus effect depends on the line. Between lines, $\Delta_{\mathrm{T47DR–MCF7R}}$ is -0.16 ± 0.09 in the cytoplasm and $+0.41 \pm 0.08$ in the nucleus; however the two lines were measured on different days, so this comparison is confounded with the session. The four confirmatory tests after Holm correction have $p = 2 \cdot 10^{-16}$, 0.021 (T47DR nucleus–cytoplasm), 0.089 and $5 \cdot 10^{-6}$.

Systematics. Nine variants change one preprocessing choice at a time (baseline fraction, threshold, lower and upper fit-window edge). With the Barlow criterion [5] no variant is significant for MCF7R and the total systematic is 0.07 for T47DR, smaller than its statistical error 0.10 (Fig. 6, right).

V. BAYESIAN ANALYSIS

Model and sampling. The same model with weakly informative priors ($\mu_g \sim \mathcal{N}(0, 1)$, $\sigma_g \sim \mathrm{HalfNormal}(1)$) is sampled with `emcee` [6] (40 walkers; $\hat{R}_{max} < 1.03$, integrated autocorrelation time $\lesssim 100$ steps). The posteriors of Δ (median, 68% interval) are $-0.81\,[-0.89, -0.74]$ and $-0.25\,[-0.34, -0.14]$, with $P(\Delta < 0) = 1.00$ and 0.993; they coincide with the profile intervals and are stable un-

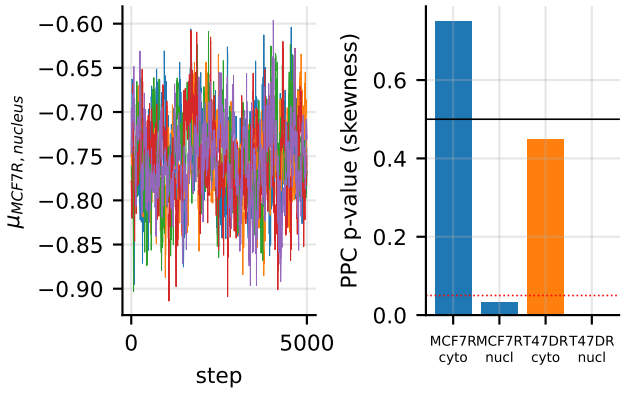


FIG. 4. Left: trace of $\mu_{MCF7R,nucleus}$ for five walkers. Right: posterior predictive p-value of the skewness in each group (dotted: 0.05).

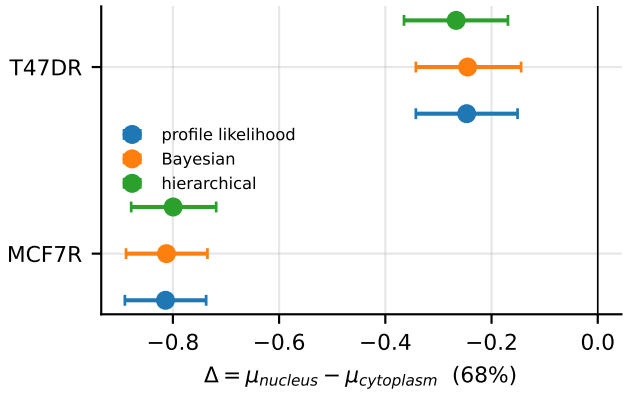


FIG. 5. Δ (nucleus – cytoplasm) in $\log A$ with the three methods.

der wide, narrow and Jeffreys priors (shifts ≤ 0.03).

Bayes factors. Marginalising σ analytically, the evidence of H_1 ($\Delta \sim \mathcal{N}(0, \tau^2)$) against H_0 ($\Delta = 0$) is computed by exact numerical integration. For MCF7R $\log_{10} BF_{10} \simeq 14$ for any $\tau \gtrsim 0.3$. For T47DR $\log_{10} BF_{10} \leq 0.8$ (about 6:1 at best, 0.6 at $\tau = 0.5$) and negative for $\tau \gtrsim 5$ (Fig. 6, left): a small effect that $p \simeq 0.01$ would call significant is only weak evidence in the Bayesian sense [7].

Model checks. A Student-t likelihood is not preferred (WAIC [8] difference -0.4, $\nu \simeq 39$), so heavy tails are not the problem. The posterior predictive check reproduces mean and standard deviation in all groups but not the *skewness* of the nucleus groups (T47DR nucleus: $p_{PPC} \simeq 0$; MCF7R nucleus: 0.04; Fig. 4). The chains are well mixed (Fig. 4, left).

Hierarchical model. Writing $\mu_{l,nucleus} = a_l + b + u_l$ with $u_l \sim \mathcal{N}(0, \tau_u^2)$, the common effect is $b = -0.49\,[-0.77, -0.16]$ and $\tau_u = 0.43\,[0.26, 0.71]$, confirming that the effect differs between lines. Partial pooling moves the line effects by less than 0.03.

Low statistics. With 5–80 curves per group, the 68% credible interval over-covers slightly (0.72 at $n = 5$, 0.68 at $n = 80$) and the profile interval under-covers (0.62 at $n = 5$, 0.66 at $n = 80$), at the price of intervals longer by 30% at $n = 5$; the two agree for $n \gtrsim 20$.

VI. MACHINE LEARNING

Anomaly detection with known truth. Since all real curves are pre-selected, the efficiency of anomaly detection is measured on 1000 synthetic Hertz-like curves whose parameters are sampled from the real fits (including a correlated model mismatch), with 5% of injected defects (`no_contact`, `saturation`, `huge_noise`, `jump`, `wrong_slope`). Isolation Forest [9] and a robust Mahalanobis distance, both unsupervised and trained only on

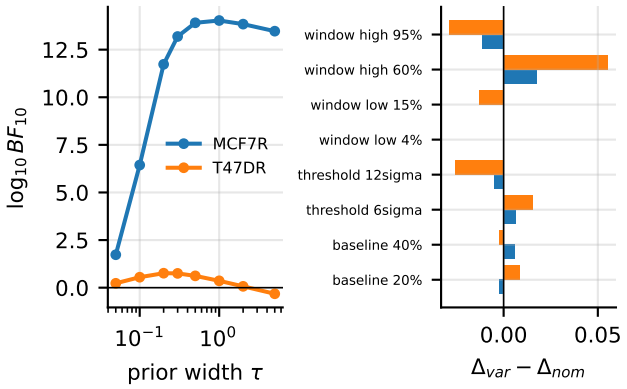


FIG. 6. Left: Bayes factor versus the prior width on Δ. Right: shift of Δ for each preprocessing variant.

good curves, reach AUC 0.88 and 0.87; at a 5% false-positive rate the efficiencies are 100% for `no_contact` and `huge_noise` (these already fail the fit), 60–90% for `saturation` and `jump`, and only 10–30% for `wrong_slope` (Fig. 8, left; 10 defects per type, i.e. about ±15%). Applied to the real curves (trained on them, cross-validated), the detector flags 8 curves in common with the second detector, for visual inspection only.

Classification of the region. From $\log A$, maximum indentation and two adhesion features, a logistic model separates nucleus and cytoplasm with AUC 0.95 in MCF7R and 0.63 in T47DR (cross-validated; permutation tests $p = 0.005$ and 0.025), consistent with the effects of the previous sections (Figs. 8, right, and 7). Trained on one line and tested on the other, the AUC is 0.66 (MCF7R→T47DR) and 0.96 (T47DR→MCF7R): the sign of the effect is shared, its size is not.

Curve shape and a model-independent check. A sixth notebook resamples every force–indentation curve on 60 points and removes the amplitude. Without labels the curves do not cluster by line (adjusted Rand index ≈ 0) and only weakly by region (≤ 0.2). Classifi-

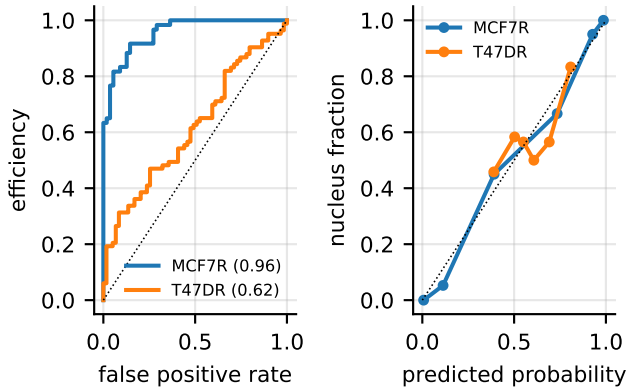


FIG. 7. Cross-validated ROC (AUC in the legend) and calibration of the logistic nucleus/cytoplasm classifier.

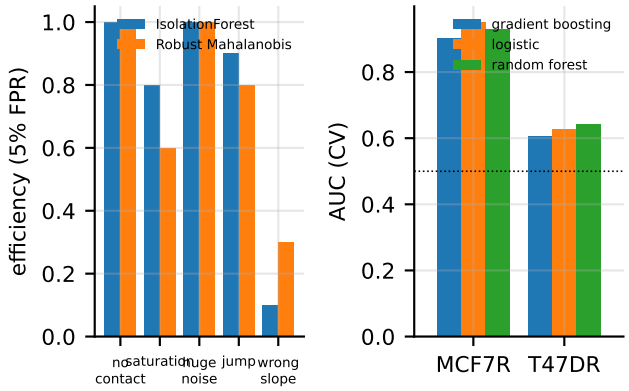


FIG. 8. Left: anomaly-detection efficiency on synthetic defects. Right: cross-validated AUC of the nucleus/cytoplasm classifiers.

cation from the shape alone gives AUC 0.62 in MCF7R (where the amplitude carries the information, 0.95) but 0.76 in T47DR, against 0.63 from the scalar features: in T47DR the shape contains region information that A does not, which needs checking on the raw curves. Because the curves end at a nearly constant setpoint force, the maximum indentation is a function of A, so the strong correlation ($\simeq -0.9$) of $\log A$ with it is expected and not an artifact. As a test independent of the contact model, the force at a fixed indentation of 300 nm is lower in the nucleus in both lines, $\Delta \log F = -0.76$ [95% interval $-0.91, -0.61$] (MCF7R) and -0.35 $[-0.53, -0.18]$ (T47DR, $p \approx 2{\cdot}10^{-4}$): the conclusion does not depend on the Hertz fit or its window, and for T47DR it is clearer than with A. Conformal prediction at 90% confidence gives a single label to 79% of the MCF7R curves but only 25% of the T47DR ones, and the learning curve saturates at AUC ≈ 0.95 after about 10 curves per class for MCF7R, while T47DR stays near 0.6.

Depth-resolved stiffness and the T47DR signal. A seventh notebook follows the nucleus–cytoplasm difference as a function of indentation. The nucleus is softer at *every* depth in both lines: in the range 150–400 nm, where almost all curves contribute, $\Delta \log F(\delta_0)$ is between -0.70 and -0.80 for MCF7R and between -0.31 and -0.37 for T47DR, with 95% intervals excluding zero; at larger depths fewer and softer curves reach the depth, so the apparent shrinkage is partly a selection effect. At a fixed force the nucleus needs a larger indentation, by a factor $\simeq 1.7$ in MCF7R ($\Delta \log \delta = +0.54$) and $\Delta \log \delta = +0.15$ to $+0.19$ in T47DR ($p = 0.004$–0.03), a second check independent of the contact model (but not of the time confounding discussed below). The effective stiffness $F/\delta_0^{3/2}$ decreases by about 0.05–0.14 per 100 nm in all groups, i.e. the response is slightly sub-Hertzian in both regions. The T47DR shape signal is not an artifact of the contact point: the shape-only AUC stays at 0.75–0.78 for contact shifts of ± 50 nm and above the permutation null ($p = 0.007$). It is carried entirely by the local exponent, whose value alone gives AUC 0.76 and whose removal brings the residual shape to 0.58: the nuclear curves of T47DR have a higher free exponent (median 1.50 against 1.43, $p \approx 5 \cdot 10^{-8}$), whereas in MCF7R the two regions are indistinguishable (1.45 and 1.44). The difference is small in size and may reflect a different contact law (nuclear geometry, thickness) or a bias of the contact-point estimate; it cannot be settled with these data.

Mixtures. A parametric-bootstrap likelihood-ratio test favours two Gaussian components only for T47DR nucleus ($p = 0.010$; 71% at $\log A = -0.64$ and 29% at 0.35); this is the origin of the skewness seen in the posterior predictive check.

VII. MEASUREMENT ORDER AND DEPENDENCE BETWEEN CURVES

The JPK files carry a timestamp for each curve, which allows two checks on the experimental design.

Region is confounded with time. In both sessions the cytoplasm was measured first (about 0–23 min) and the nucleus afterwards (about 25–47 min), with no overlap. Any drift of $\log A$ during the session (wear of the tip, state of the cells, setpoint) is therefore added to the region difference. Regressing $\log A$ on region and on a common within-session time trend, with standard errors clustered on the bursts defined below, gives the difference *at a common time*: for MCF7R the within-block slope is positive ($+0.06 \pm 0.06$ per 10 min), opposite to the effect, and the difference becomes $\Delta = -0.97 \pm 0.17$. For T47DR $\log A$ decreases during the session (-0.13 ± 0.07 per 10 min); projected over the ≈ 22 min between the block centres this drift alone predicts $\Delta \approx -0.30$, i.e. the entire observed value (-0.25), and the difference at a common time is $+0.05 \pm 0.17$, compatible both with zero and with the observed value. The force at a fixed indentation of 300 nm behaves in the same way (T47DR: slope

-0.16 per 10 min, $p = 0.014$; difference at common time 0.01 ± 0.16). The free exponent is different: it shows no drift in T47DR ($p = 0.52$), and the excess of the nucleus survives the correction ($+0.059 \pm 0.017$, $p = 0.002$). Thus the stiffness difference of T47DR cannot be distinguished from a temporal drift, whereas the difference in the *form* of the response can.

Bursts of curves as a proxy of the cell. The gaps between consecutive curves are bimodal, about 2 s (the next curve at the same spot) against 60–100 s (a new location). A gap above 10 s defines a burst; 13–17 bursts are found per group, of 3–5 curves each, stable for any threshold between 5 and 60 s. The intra-class correlation of $\log A$ is moderate (0.10–0.18 in three groups, ≈ 0 in MCF7R nucleus), the design effect is about 1.4 and the effective sample size is 38–73 curves instead of 55–83. Pseudo-replication is therefore real but mild: the standard error of Δ increases by 5% (MCF7R) and 16% (T47DR). At the burst level MCF7R is unchanged ($p < 10^{-3}$ with every method), while for T47DR the naive $p = 0.011$ becomes 0.035 (cluster-robust regression), 0.049 (cluster permutation) and 0.095 (Welch on burst means), with a cluster-bootstrap interval for Δ of $[-0.46, -0.03]$. Several bursts may belong to the same cell, so these are lower bounds on the dependence.

VIII. CALIBRATION, REPEATABILITY AND A UNIFIED MODEL

Calibration between sessions. Both sessions used the same cantilever (MLCT-BIO, id 2300FE), but the stored spring constant is 0.119 N/m on 25/02 and 0.080 N/m on 18/03 (ln ratio -0.40), and the sensitivity 10.9 and 14.0 nm/V; the force setpoint is the same, 0.75 nN. Since $F \propto k$, a calibration uncertainty of this size shifts $\ln A$ by up to 0.4, as large as the observed differences between lines (-0.16 in the cytoplasm, $+0.41$ in the nucleus; $+0.24$ and $+0.80$ if both sessions had the same k): *differences between lines cannot be interpreted, not even in sign.* Within a session the calibration is common to both regions and cancels in the nucleus–cytoplasm difference.

Repeatability and preconditioning. Curves taken a few seconds apart at the same spot differ by sd = 0.33–0.61 in $\log A$ (a factor 1.4–1.8), 80–100% of the total variance: the scatter at a single spot is as large as that between spots. Moreover the first curve at a spot is stiffer than the following ones: the paired difference second − first is -0.25 ± 0.09 ($p = 0.013$, 61 bursts), with the same sign in all four groups, so the indentation modifies the cell. Since the nucleus bursts are longer (4.6–5.2 curves against 3.4–3.5), the nucleus mean contains more softened curves. Using only the first curve of each burst, the difference is $\Delta = -0.77 \pm 0.13$ for MCF7R (unchanged) and -0.01 ± 0.26 for T47DR, compatible both with no effect and with the earlier value because only 33 curves remain.

A single hierarchical model. The three issues are combined in one model per line, $y_i = a + \beta\, r_i + \gamma\, t_i + \delta_{p_i} + u_b + \varepsilon_i$, with r the region, t the time, δ_p the position in the burst and $u_b \sim \mathcal{N}(0, \tau^2)$ a burst random effect integrated out analytically; it is sampled with `emcee`. For MCF7R $\beta = -0.79$ with the region alone and -0.88 with drift and position, 95% interval $[-1.22, -0.55]$, $P(\beta < 0) = 1.00$ for any prior width of the drift. For T47DR $\beta = -0.22$ $[-0.45, +0.02]$ ($P = 0.97$) already with the bursts alone, and $+0.03$ $[-0.35, +0.41]$ ($P = 0.43$) with the drift term; the data estimate a negative drift (≈ -0.1 in $\log A$ per 10 min). If one admits only a very small drift ($s_\gamma \leq 0.05$ sd per 10 min) $P(\beta < 0)$ is 0.85–0.92, never above; for a wide prior it falls to 0.34–0.42: the T47DR stiffness difference is not established, and depends entirely on assuming a negligible drift. The excess of the free exponent in the T47DR nucleus is robust ($+0.062$ $[0.019, 0.103]$ with drift and position; $P(\beta < 0) = 0.003$), whereas for MCF7R it is not significant ($+0.036$ $[-0.018, 0.089]$). The four models are only weakly distinguished by the cluster-level WAIC (differences of 2–3 units), as expected from the weak identifiability of drift and region when the two regions occupy separate time blocks. The difference of the two region effects, free of the calibration error because each β is a within-session difference but still confounded with the session, is $\beta_{\mathrm{MCF7R}} - \beta_{\mathrm{T47DR}} = -0.93$ $[-1.42, -0.43]$.

IX. DISCUSSION AND LIMITS

The three approaches agree on the size of the effect and differ only in how they weigh the evidence for T47DR, which is the honest summary: the nucleus of MCF7R is clearly softer than its cytoplasm ($\times 0.44$ in A), while for T47DR the stiffness effect ($\times 0.78$ in the naive analysis) is not established: it is below the 80%-power threshold, weakens at the burst level, vanishes if only the first curve of each burst is used, and is compatible with a temporal drift because the two regions were measured in consecutive blocks. What is robust for T47DR is the higher contact-law exponent of the nucleus. The main limits are: (i) **pseudo-replication**: several curves come from the same cell and the cell identifier is not available; the bursts of consecutive curves show that the errors are optimistic by about 5–16%, as a lower bound, and a hierarchical model with the cell as grouping factor is the natural extension; (ii) **cell line ≡ session and region ≡ time block**: the lines were measured on different days (25/02 and 18/03), so differences between lines cannot be separated from session effects, and within each session the two regions were measured one after the other, so the region difference cannot be separated from a drift except through the within-block trend; the spring constant stored for the same cantilever also changes by 40% between the sessions, which makes differences between lines uninterpretable; measuring the two regions *alternately* on the same cells, and recording the cell identifier,

would settle both points; (iii) **pre-selected sample**: no real defects are available, so the efficiency of the anomaly detection is optimistic; (iv) **effective parameter**: A is not the Young modulus and assumes a 3/2 exponent; (v) the high-A tail of T47DR nucleus suggests subpopulations, which a comparison of means does not describe.

X. CODE

All results are produced by ten notebooks in `Code_General/` (`01` reading and QC, `02` contact point and fit, `03` frequentist, `04` Bayesian, `05` machine learning, `06` curve-shape extensions, `07` depth-resolved stiffness, `08` acquisition time and clustering, `09` calibration and repeatability, `10` unified model) and the shared module `afm_tools.py`; the figures are regenerated by `Docs/report/make_figures.py` from `Code_General/results/`. The raw data are not distributed with the repository (see the README).

Appendix A: Notation and formulas

Quantities. A: effective elastic parameter ($\mathrm{nN}\,\mu\mathrm{m}^{-3/2}$); $x = \log A$; $\mu_{l,r}, \sigma_{l,r}$: mean and standard deviation of x for line $l \in \{\mathrm{MCF7R}, \mathrm{T47DR}\}$ and region $r \in \{\text{cytoplasm, nucleus}\}$; $\Delta_l = \mu_{l,nucleus} - \mu_{l,cytoplasm}$; e^{Δ}: ratio of typical A.

Profile likelihood. With n_c, n_n curves and $S_c(\mu) = \sum_i (x_i - \mu)^2$ over the cytoplasm curves, $S_n(\mu, \Delta) = \sum_j (x_j - \mu - \Delta)^2$ over the nucleus curves, profiling $\sigma_c^2 = S_c/n_c$ and $\sigma_n^2 = S_n/n_n$ gives

$$-2 \ln \mathcal{L}_{\mathrm{prof}}(\Delta) = \min_{\mu} \left[n_c \ln \frac{S_c(\mu)}{n_c} + n_n \ln \frac{S_n(\mu, \Delta)}{n_n} \right] + \text{const.}, \tag{A1}$$

and the 68% interval is where this rises by one unit above its minimum; the test of $\Delta = 0$ uses $q = -2 \ln \mathcal{L}_{\mathrm{prof}}(0) + 2 \ln \mathcal{L}_{\mathrm{prof}}(\hat{\Delta}) \sim \chi^2_1$.

Bayes factor. With the improper prior $p(\sigma) \propto 1/\sigma$ common to both hypotheses, integrating σ_c, σ_n out gives

$$p(\text{data} \mid \mu, \Delta) \propto S_c(\mu)^{-n_c/2}\, S_n(\mu, \Delta)^{-n_n/2}, \tag{A2}$$

and $BF_{10} = \iint p(\text{data} \mid \mu, \Delta)\, \mathcal{N}(\mu; 0, 1)\, \mathcal{N}(\Delta; 0, \tau^2)\, d\mu\, d\Delta \,/ \int p(\text{data} \mid \mu, 0)\, \mathcal{N}(\mu; 0, 1)\, d\mu$, evaluated on a grid.

Barlow criterion. A variant with estimates Δ_v and error σ_v gives a significant shift if $|\Delta_v - \Delta_{nom}| / \sqrt{|\sigma_v^2 - \sigma_{nom}^2|} > 2$.

Anomaly score and efficiency. The anomaly score is minus the Isolation Forest score (or the robust Mahalanobis distance); the threshold is the 95th percentile of the cross-validated scores of good training curves, so the false-positive rate is 5% by construction; the efficiency of a defect type is the fraction of defective curves above threshold, counting failed fits as detected.

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